

MOS Transistors Final Project Report

Kaden Du and James Jagielski

Introduction

To create an accurate model of the provided MOSFET data, we used the EKV model, given by:

$$(1) I_D = I_S \log^2(1 + e^{(\kappa(V_{GB} - V_T - V_{SB}))/2U_T}) - I_S \log^2(1 + e^{(\kappa(V_{GB} - V_T - V_{DB}))/2U_T})$$

In this model, κ is a constant value, fit to $\partial\psi_s/\partial V_G$. U_T denotes the thermal voltage kT/q . I_S is denoted by the equation:

$$(2) I_S = 2 u_{eff} C_{ox} \frac{W}{L} (U_T^2) / \kappa$$

Short channel effects will also be added as a factor to I_S , and will be described later on.

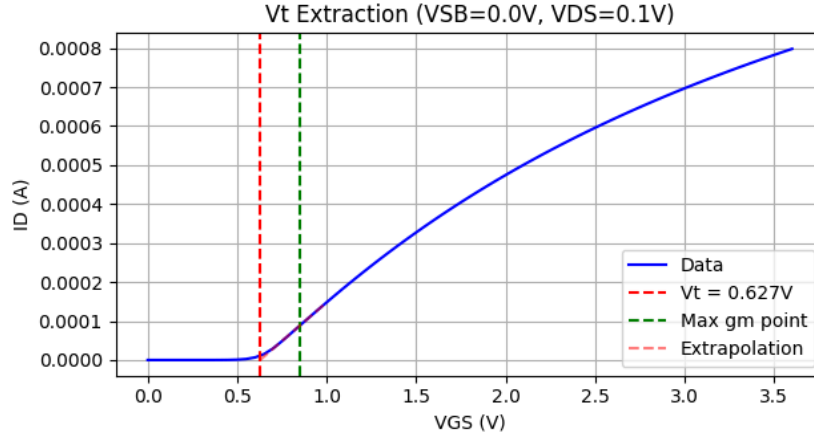
Creating the Model

Fitting Threshold Voltage

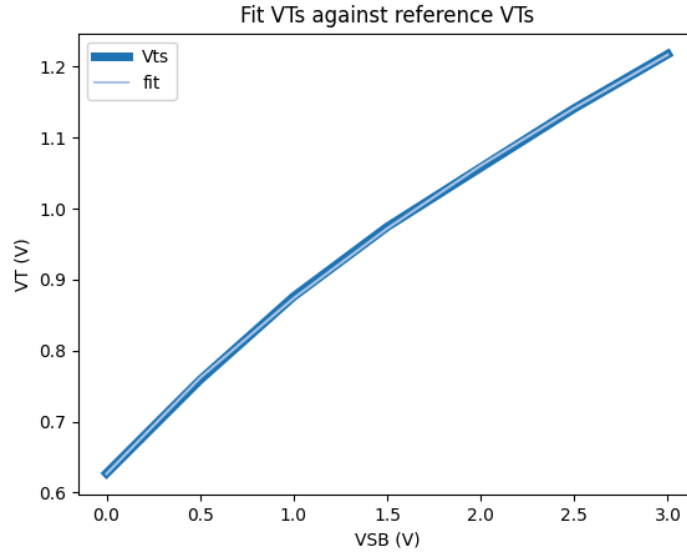
The first parameter to be fit is the threshold voltage and its corresponding parameters. V_{th} is given by:

$$(3) V_{th} = V_{FB} + \Phi_0 + \gamma \sqrt{\Phi_0 + V_{SB}}$$

First, V_{th} values were extracted from the I_D - V_{GS} data by extrapolating a linear fit of the curve in moderate-strong inversion, and finding the x intercept. An example of one of these fits is shown below:



V_T is fitted for all V_{SB} values with $V_{DS} = 0.1$ V. Thus, a plot of V_T / V_{SB} is created. The Φ_0 value is found by minimizing the sum-squared error between $\sqrt{V_{sb} + \Phi_0}$ and the V_T array. γ is found by normalizing this plot to $\sqrt{V_{sb} + \Phi_0}$, and finding the slope of the new graph. Lastly, V_{FB} is trivial to solve as all other parameters in the equation are known. The fit of these parameters to measured V_T values is shown below:

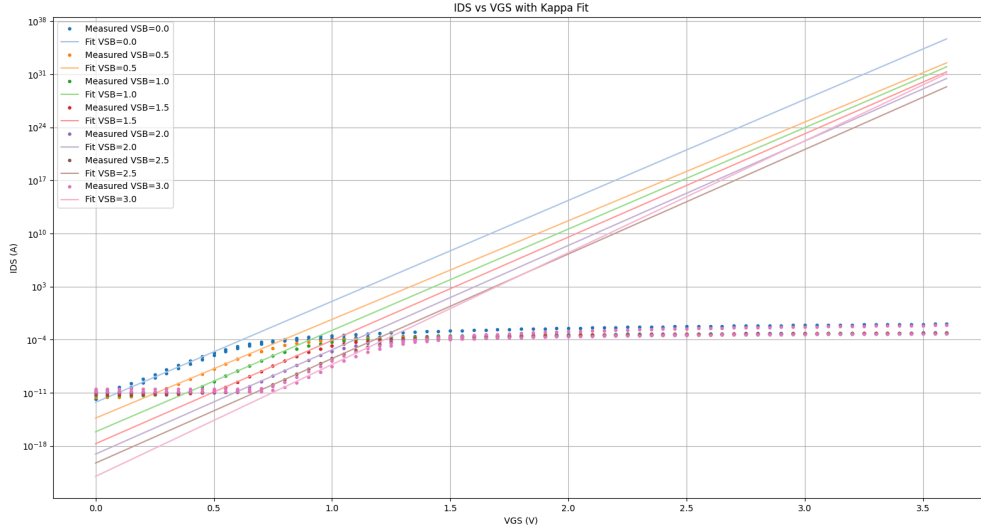


Fitting Kappa

Kappa is a unitless term that represents the capacitive divider between the oxide capacitance above the channel and the effective capacitance of the depletion layer beneath the channel. Because the thickness of the depletion layer beneath the channel increases with increasing gate voltage, C_{dep} will get smaller for larger values of V_G , which implies that κ will increase with increasing V_G .

$$(4) \kappa \equiv \partial\psi_s / \partial V_G = C_{ox} / (C_{ox} + C_{dep})$$

We fit kappa by plotting V_{gs} vs I_{ds} . This allowed us to simplify our EKV model to $ID = I_0 * e^{\kappa Vg/U_T}$ and linear fit the data for kappa and I_0 on a semi-log-y plot. The slope relates to the kappa and the intercept relates to the I_0 term.



Once we fit our data we averaged out the kappa values for one consistent value to use for the model.

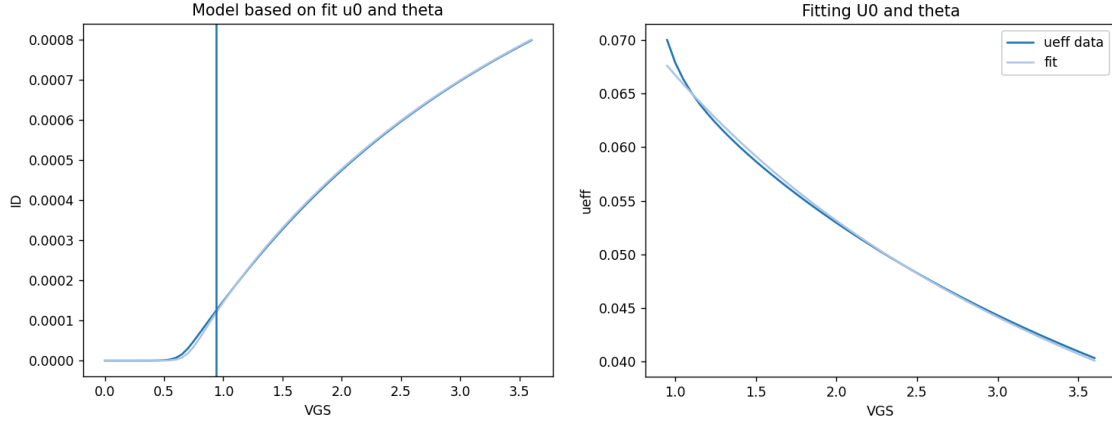
Fitting Is

I_s , as given, is a function consisting of u_{eff} and constants. Thus, to fit I_s , we must fit u_{eff} . The top level approach was to, at $V_{DS} = 0.1$ V, and $V_{SB} = 0$ V, directly calculate an array of what I_s should be to match the measured current, also creating the corresponding array of u_{eff} values. U_{eff} is specified by:

$$(5) u_{eff} = \frac{u_0}{1 + \Theta(V_{gs} - V_T) + \Theta_B V_{SB}}$$

Fitting u_0 and theta

U_0 and Θ are fit simultaneously. An array of u_{eff} s is first pulled from the measured data (from a test with $V_{SB} = 0$ to ensure no Θ_B dependance). The data is filtered to only the saturation region to simplify the curve_fitting process. Using previously determined V_T values, Θ and u_0 are fit to minimize error between the measured u_{eff} values and the newly calculated ones. To visually verify this fit, this fit is plotted, shown below:



Note that the vertical line on the ID/VGS plot is the minimum VGS value where u_0 and θ were fitted.

Fitting θ_B

θ_B is fitted similarly as θ and u_0 . An array of u_{effs} in terms of V_{SB} is pulled from the V_{GS} - I_D data, using the same V_{gs} (3.4) for each V_{SB} . θ_B is fitted to this data using (4).

Velocity Saturation and Channel Length Modulation

Lastly, the effects of both Velocity Saturation and Channel Length Modulation are taken into account. As both effects become more evident at high V_{DS} values, it was difficult to isolate one parameter to fit. So, functions to fit both parameters could be run repeatedly until parameters were converged upon for the best fit.

Velocity Saturation

The equation denoted for velocity saturation is given by:

$$(6) \quad u_{eff} = \frac{u_{eff0}}{1 + \frac{u_{eff0} E}{v_{sat}}}$$

Where E is the electric field (V_{ds} / L) and v_{sat} is the saturation velocity. As most other parameters are fit, and our results are fairly accurate at this point, we can use the full EKV model equation (1) to fit v_{sat} onto the final plot. We used $V_{Gs} = 3.4$ and $V_{SB} = 0$ to fit v_{sat} , as the high gate voltage means the waveform is clearly in velocity saturation.

Channel Length Modulation

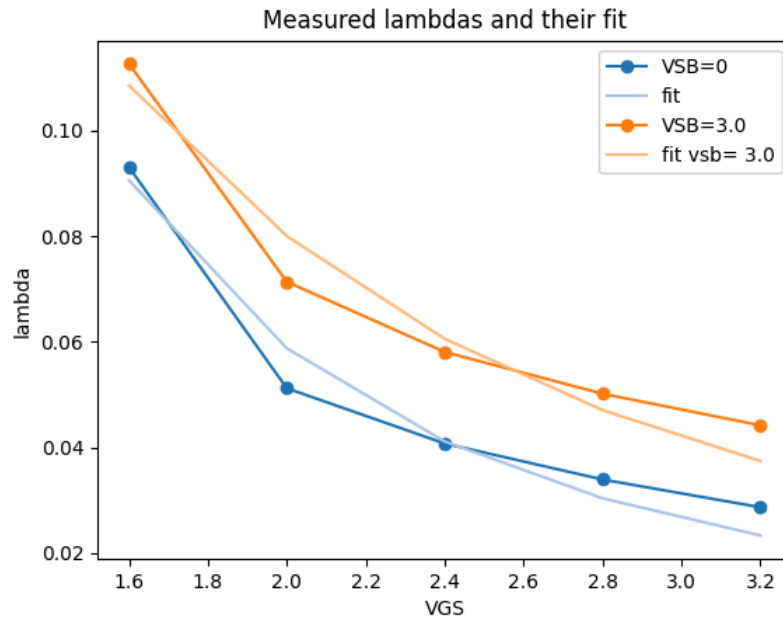
The effect of channel length modulation is given by:

$$(7) I_D = I_{D0}(1 + \lambda V_{DS})$$

Values for λ were computed using the already fit model parameters across V_{GS} and V_{SB} . By plotting these values, it was evident that λ had a large dependence on bias conditions. Through experimentation, it was found that the following equation could be used to loosely model λ 's relationship with V_{GS} and V_{SB} , although this equation does not have a physical meaning.

$$(8) \lambda = A * \frac{1 + B * V_{SB}}{1 + C * V_{GS}^2 + D * V_{SB}}$$

First, the values of A and C can be fit using data where $V_{SB} = 0$. Then, once those values were computed, the values of B and D could be fit to match the V_{SB} dependence.



Model Results

The results of our EKV model fitting are shown below. The model takes ~2 seconds to fit data and once fitted, takes 1uS per ID value returned.

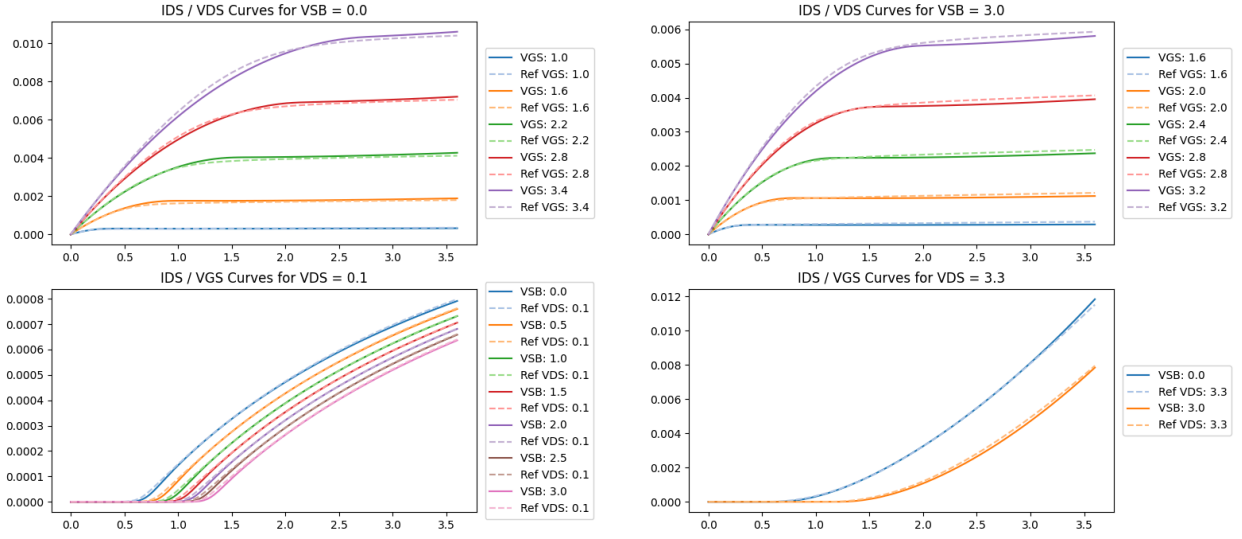


Figure: Model results plotted against input data

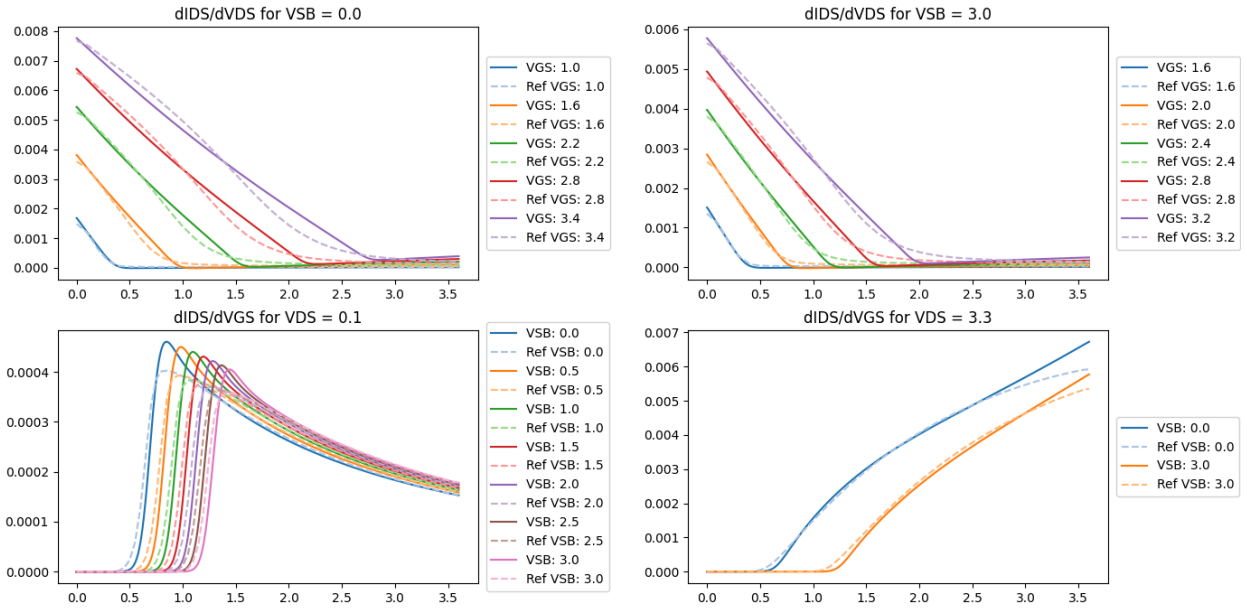


Figure: Model Derivatives against input

Table: RMS Errors for each input plot

Category	VGS	VDS	VSB	E_rms
ID/VDS	1	—	0	0.06031153
ID/VDS	1.6	—	0	0.04643605
ID/VDS	2.2	—	0	0.02694574

ID/VDS	2.8	—	0	0.01938397
ID/VDS	3.4	—	0	0.02435714
ID/VDS	1.6	—	3	0.221375
ID/VDS	2	—	3	0.172501
ID/VDS	2.4	—	3	0.09511898
ID/VDS	2.8	—	3	0.06000056
ID/VDS	3.2	—	3	0.04425473
ID/VGS	—	0.1	0	0.3578368
ID/VGS	—	0.1	0.5	0.3448961
ID/VGS	—	0.1	1	0.3838217
ID/VGS	—	0.1	1.5	0.4677908
ID/VGS	—	0.1	2	0.4901559
ID/VGS	—	0.1	2.5	0.5109531
ID/VGS	—	0.1	3	0.5303575
ID/VGS	—	3.3	0	0.3781533
ID/VGS	—	3.3	3	0.4116304
dID/dVDS	1	—	0	0.7305431
dID/dVDS	1.6	—	0	0.5841985
dID/dVDS	2.2	—	0	0.538278
dID/dVDS	2.8	—	0	0.4114776
dID/dVDS	3.4	—	0	0.31278
dID/dVDS	1.6	—	3	0.8249025
dID/dVDS	2	—	3	0.6065859
dID/dVDS	2.4	—	3	0.4590024
dID/dVDS	2.8	—	3	0.3749508
dID/dVDS	3.2	—	3	0.3172275
dID/dVGS	—	0.1	0	0.3658973
dID/dVGS	—	0.1	0.5	0.3534326
dID/dVGS	—	0.1	1	0.323935
dID/dVGS	—	0.1	1.5	0.3053477
dID/dVGS	—	0.1	2	0.2892663
dID/dVGS	—	0.1	2.5	0.2764995
dID/dVGS	—	0.1	3	0.268023
dID/dVGS	—	3.3	0	0.3998858
dID/dVGS	—	3.3	3	0.3799461

As shown, the model reports high accuracy in modeling drain current in strong inversion. As shown in the reported error values, a higher accuracy could be achieved in the weak inversion region. As most of the data is in strong inversion, this region was prioritized to get the best overall fit, sacrificing the details in weak inversion. Namingly, the parameters for channel length modulation and velocity saturation were fit while prioritizing the reduction in overall error (magnitude), not considering introduced errors as a percentage, which can disproportionately affect smaller current regions. Moving forward, it would be better to fit these parameters with more physical meaning behind their measurements. For example, measuring the slope of I_d to solve for λ . Note that this was attempted during our work, but due to the contradicting dependence on λ and v_{dsat} in the same region, we were not able to get a channel length modulation fit that gave accurate results. Alternatively, we could use a different fitting function with the same approach, instead minimizing percent error rather than total error.

Secondly, the derivatives of our model were not as accurate as they could have been. To fix these in modeling moving forward, we could similarly design a fitting function which would account for both the error in the function, and the error in the derivative, allowing us to fit our parameters, such as θ or anything else with dependence on V_{GB} , more accurately.

Tests

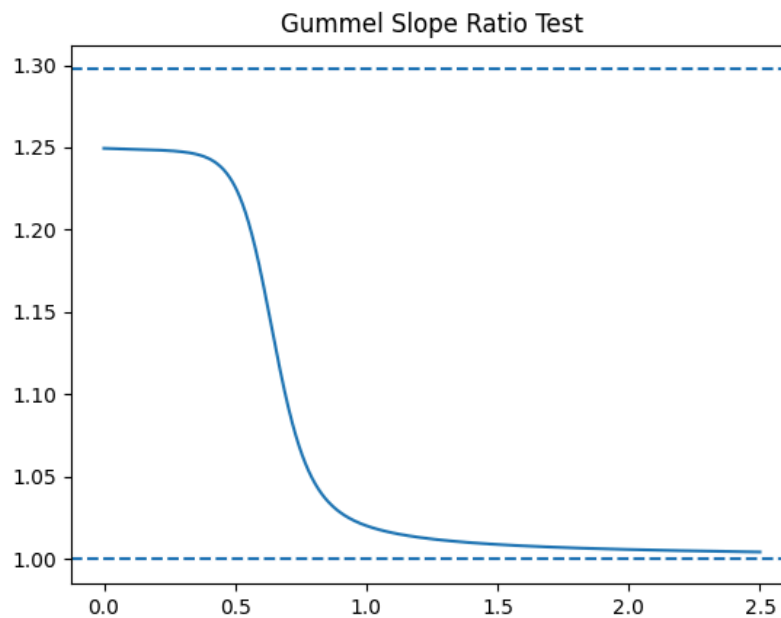
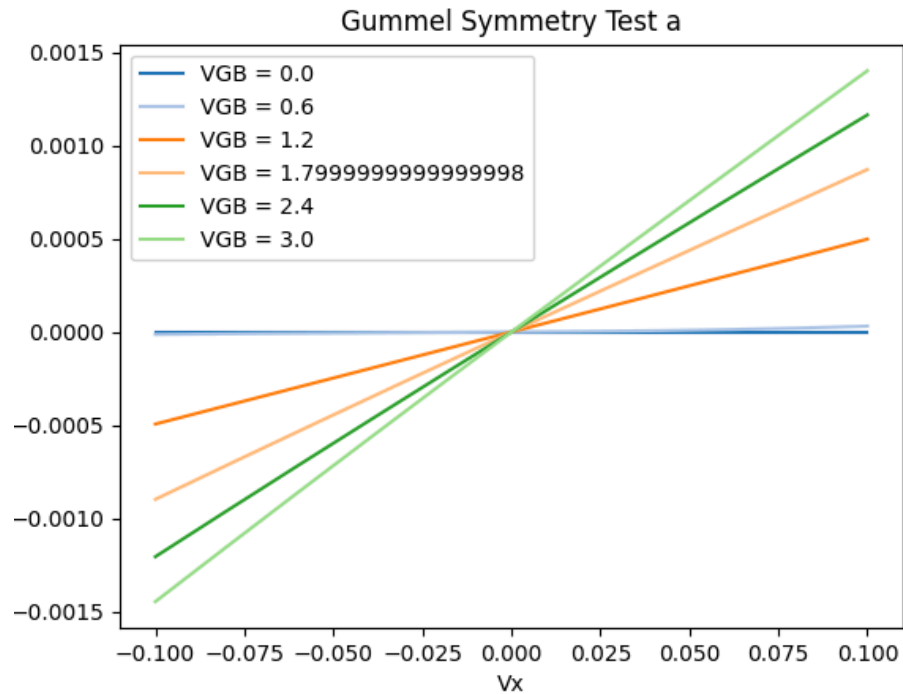
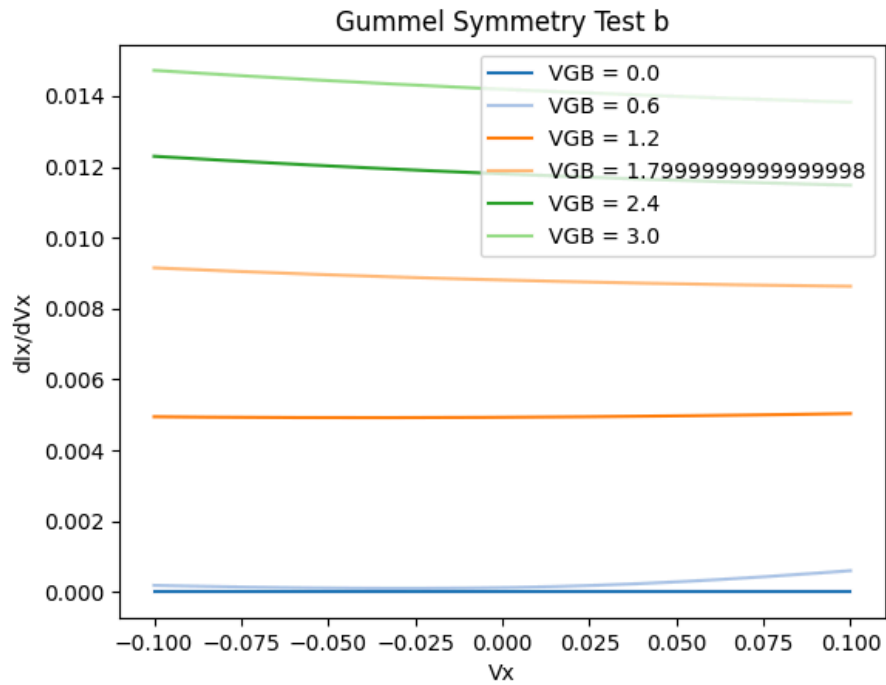


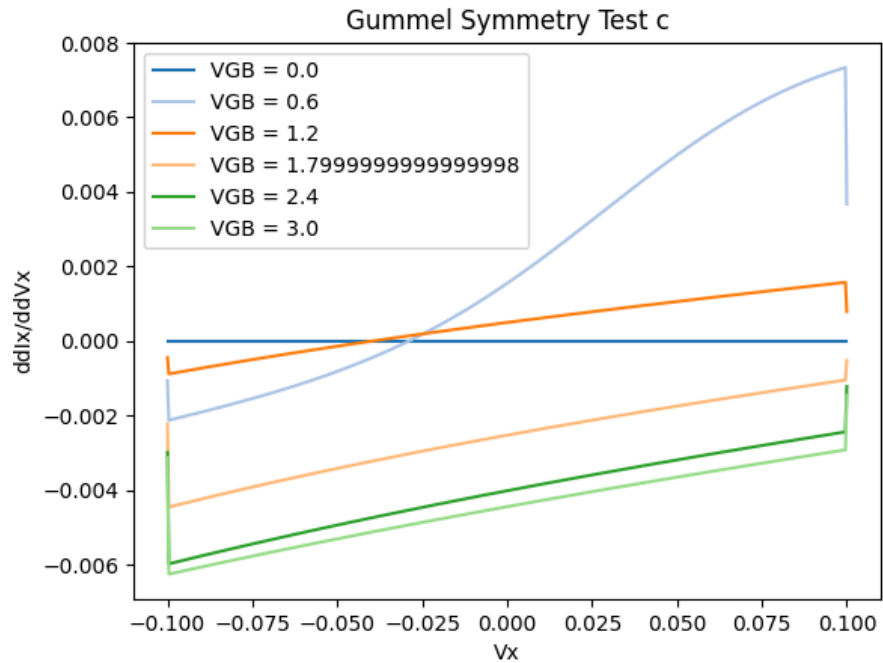
Figure: Gummel Slope Ratio Test at $V_{ds} = 0.5 \phi T$

When tested using the gummel slope ratio test, our model does not saturate to $1.30 S_R$ when close to 0 V_{in} . As described by the test, this indicates, this indicates that this model is not accurate in weak inversion. This is further backed up by the errors in our VGS-ID plots, which report high RMS errors, even when visually matching the waveform.

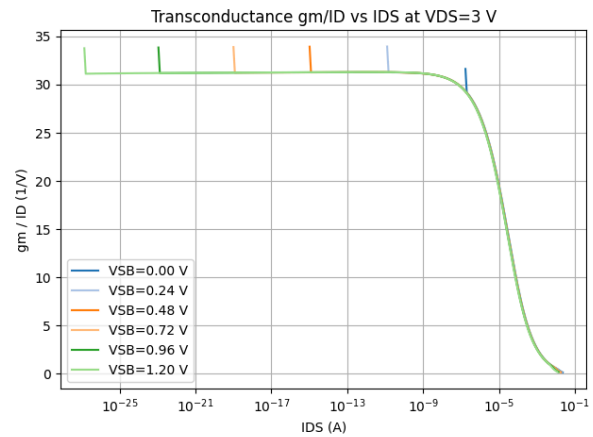
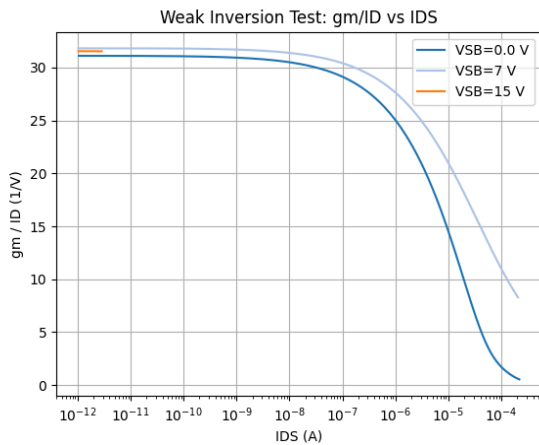


We moved to the K2 testing procedures in the textbook appendix. The gate and substrate currents are contained in equal amounts in I_d and I_s , at $V_{DS} = 0$ and therefore cancel out in the difference. This makes I_x an odd function of V_x . Plotting I_x , dI_x/dV_x , and d^2I_x/dV_x^2 as a function of V_x allows the symmetry to be visualized. Our plots show the symmetry, but in the first derivative there's not as much slope as the example plots.

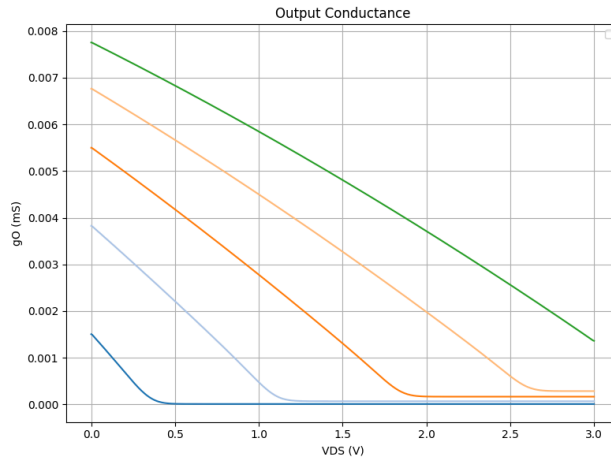




After analyzing the gummel symmetry tests we can see our model runs smoothly. We show we have no asymptotes through the same range. Our values for the derivatives are much smaller. We believe this is caused by the smaller device sizing causing the current level to stay lower.



After much troubleshooting, we got the g_m/i_d curve to separate at 7V. This is very inaccurate, but in the right graph we can see the graph follows the correct form, but it doesn't have any V_{sb} variation.



The last test we ran was the transition from nonsaturation to saturation. We expected a curved approach to the x-axis, but we have a sharp transition. This means our model includes "if" conditions at the transition from nonsaturation to saturation; they predict a continuous g_0 but produce physically incorrect shapes. The EKV model has a smoothing function built into it with the forward and reverse current, so this transition makes sense to be a sharp transition, but we thought this transition would be much smoother.

Conclusion

Overall, the EKV model is a powerful model that is simplistic for modeling by hand. The model can easily be expanded for short channel effects and other physics based variations accommodating them into a neat physics term I_s . The model breaks down into its respective modes of inversion and ohmic versus saturation neatly for hand calculations. We struggled with the velocity saturation and early effect implementations. We had slight variations in the IV-characteristics, which are believed to be caused by imperfect early effect and velocity saturation. Based on the tests and how closely our IV-curves were, we believe this is a fully accurate model.

Code Instructions

The full model can be run from `main.py`, which will load data, fit all parameters, plot results, and run all tests. Some of the fitting parameters will only plot results, if the `plot=True` argument is passed to them, in which case the code will need to be edited. All the code for the model is available in `model.py`. These were omitted because there are far too many plots to be convenient for the user to see all of them at any given time. The code depends on having libraries `numpy`, `matplotlib`, and `scipy` installed to run.

The zipped code repository is attached on Canvas, but is also available here:

<https://github.com/dukad/MOSTransistorsFinalProject>